

Problem Set 5.1 – Multi-Agent Intelligence Hub

Kshitiz Reddy
M.Tech AIML, School of AI, Bennett University

Source Code

```
1  #.setdefault is right for the Movement Log because an agent might not have
2  # a key in the log yet on its first move..setdefault(agent_id, []) inserts
3  # an empty list only if it's missing, then returns the list either way, so
4  # the same append line works for the first move and every move after.
5  #
6  # A tuple is right for Location because coordinates shouldn't change once
7  # set - if you want a different position you're really describing a new
8  # state, not editing the old one. Immutability guarantees nothing else in
9  # the program can silently change an agent's position from under you.
10 #
11 # Author: Kshitiz
12
13
14 def find_common_intelligence(registry):
15     knowledge_sets = [agent_data["Knowledge"] for agent_data in registry.values()]
16     return set.intersection(*knowledge_sets)
17
18
19 def relocate_agent(registry, log, agent_id, new_location):
20     print(f"Attempting direct mutation of {agent_id}'s Location tuple...")
21     try:
22         registry[agent_id]["Location"][0] = 0
23     except TypeError as e:
24         print(f"TypeError caught: {e}")
25         print("Tuples are immutable; reassigning a new tuple instead.")
26
27     registry[agent_id]["Location"] = new_location
28     log.setdefault(agent_id, []).append(new_location)
29
30     print(f"{agent_id} new state: {registry[agent_id]}")
31     print(f"Movement Log: {log}")
32
33
34 agent_registry = {
35     "Agent_Alpha": {
36         "Location": (10, 20),
37         "Knowledge": {"grid_map", "comm_protocol", "telemetry_sync"},
38     },
39     "Agent_Beta": {
40         "Location": (15, 5),
41         "Knowledge": {"grid_map", "comm_protocol", "power_grid"},
42     },
43     "Agent_Gamma": {
44         "Location": (0, 0),
45         "Knowledge": {"grid_map", "comm_protocol", "power_grid", "telemetry_sync"},
46     },
47 }
48
49 movement_log = {}
50
51 print("--- Common Intelligence Across All Agents ---")
```

```

52 print(find_common_intelligence(agent_registry))
53 print()
54
55 print("--- Relocating Agent_Alpha ---")
56 relocate_agent(agent_registry, movement_log, "Agent_Alpha", (12, 22))
57 print()
58
59 print("--- Relocating Agent_Alpha again ---")
60 relocate_agent(agent_registry, movement_log, "Agent_Alpha", (14, 25))
61 print()
62
63 summary_report = {
64     agent_id: len(data["Knowledge"]) for agent_id, data in agent_registry.items()
65 }
66 print("--- Summary Report (Dictionary Comprehension) ---")
67 print(summary_report)
68
69 # Submission Note: the TypeError says a tuple "does not support item
70 # assignment" - meaning the slots inside a tuple aren't writable after
71 # creation. A list stores pointers to objects in a mutable array, so
72 # list[0] = x just swaps what that slot points to. A tuple's slots are
73 # fixed at creation time, so there's no operation in the type that lets
74 # you rewrite one - the only way to get a "different" tuple is to build
75 # a new one and rebind the name to it, which is exactly what
76 # relocate_agent does after the failed mutation attempt.

```

Trace Log

```

1  --- Common Intelligence Across All Agents ---
2  {'comm_protocol', 'grid_map'}
3
4  --- Relocating Agent_Alpha ---
5  Attempting direct mutation of Agent_Alpha's Location tuple...
6  TypeError caught: 'tuple' object does not support item assignment
7  Tuples are immutable; reassigning a new tuple instead.
8  Agent_Alpha new state: {'Location': (12, 22), 'Knowledge': {'comm_protocol',
9  'grid_map', 'telemetry_sync'}}
10 Movement Log: {'Agent_Alpha': [(12, 22)]}
11
12 --- Relocating Agent_Alpha again ---
13 Attempting direct mutation of Agent_Alpha's Location tuple...
14 TypeError caught: 'tuple' object does not support item assignment
15 Tuples are immutable; reassigning a new tuple instead.
16 Agent_Alpha new state: {'Location': (14, 25), 'Knowledge': {'comm_protocol',
17 'grid_map', 'telemetry_sync'}}
18 Movement Log: {'Agent_Alpha': [(12, 22), (14, 25)]}
19
20 --- Summary Report (Dictionary Comprehension) ---
21 {'Agent_Alpha': 3, 'Agent_Beta': 3, 'Agent_Gamma': 4}

```

Common intelligence, movement log, and summary report all match the worked trace in the assignment (set element order differs, which is expected since Python does not guarantee set ordering).

Submission Note

The `TypeError` says a tuple “does not support item assignment” – its slots are fixed at creation and can’t be rewritten. A list stores pointers in a mutable array, so `list[0] = x` just swaps what that slot points to. A tuple has no such operation, so the only way to get a different value

is to build a new tuple and rebind the name to it, which is exactly what `relocate_agent` does after the failed mutation attempt.